

# Kolagani Pavan Kalyan

kolagani.pavankalyan2003@gmail.com | +91 8985229562 | github.com/Cintu07 | linkedin.com/in/pavankalyan-kolagani

## EXPERIENCE

### Founding Engineer

Nov 2025 – Aug 2026

*Cortex Technology (cor-tex.solutions), voice agents for restaurants and clinics*

*Remote*

- Callers were being cut off mid sentence, because one turn-detection profile was serving both the greeting and the conversation. Split it in two, a 0.5 threshold with 450 ms of trailing silence once the call is running and a stricter 0.8 with 700 ms during the greeting, so line noise on pickup stopped registering as speech and the agent held its turn until someone actually spoke.
- The agent sounded like it was calling from an empty room, which callers read as a robot. Built an ambience mixer that blends a restaurant bed into the outbound stream at 20 ms mu-law frames with crossfade, level-clamped so it never masks the speech underneath. Also cut model latency on live calls by moving to smaller speech models, then reverted when word accuracy dropped.

### Technical Partner and Shareholder

Jul 2026 – Present

*Dhanam Collections (dhanamcollections.com), handloom retail*

*Andhra Pradesh*

- Hosting was climbing on 12.4M serverless invocations a month for a shop with 45 products. Traced it to link prefetch: every product tile entering the viewport was server-rendering a page nobody opened. Disabled prefetch on the dynamic routes and the bill fell 72%, \$87 a month to \$25.
- Only engineer on the system: 285 commits in 43 days, 24,676 lines of TypeScript, 43 page routes, 38 API routes, 20 Postgres tables, live card payments. INR 5 lakh in sales across 200+ orders to date, on 32,376 visitors and 299,734 page views in the first week at a 12% bounce rate, with function errors under 0.1% and no timeouts through the order spikes.

### Co-Founder

Jun 2026 – Sep 2026

*OnePurplePen (onepurplepen.com), building OneDB*

*Remote*

- Built the directory engine behind OneDB: per-item slug pages, an MCP directory, per-directory titles and descriptions, and icon upload, so a new collection could be listed without shipping a deploy for it.
- Search crawlers could read the listings but agents could not. Wrote the machine-readable surface, `11ms.txt` plus JSON-LD structured data on every item, so an LLM crawler resolves a listing to structured fields instead of scraping the rendered page.

## OPEN SOURCE (RUST)

### apache/arrow-rs

Rust | 3.6k stars | columnar memory format

*1 merged, 1 open, 2 issues filed*

*Sep 2026*

- `concat` on `FixedSizeBinary` narrowed the combined length through an unchecked `as i32`, so a caller-controlled element width wrapped negative. Swapped in `i32::try_from`, the fallible pattern already used directly above it. Filed and merged the same day (#10981).
- `substring` wraps start and length to `i32` on `Utf8` but not on `LargeUtf8`. Same call, same data, different answer depending on which offset width the array happens to use (#10983).

### HelixDB/helix-db

Rust | 5.9k stars | graph + vector database

*8 merged, 1 open*

*Jul 2026 – Sep 2026*

- The NEON and scalar distance kernels disagree on 38% of random vector pairs at 22 dimensions and 93% at 1536, worst case 30 ulp. The equivalence test guarding them fed integers, where every intermediate is exactly representable, so it agreed bit for bit and could not fail. Added one that can, and checked that by tightening its tolerance until it broke (#1075).
- Proved a Louvain modularity invariant test could not fail: injected one bug, then made the old whole-phase test pass and a new per-move test fail on the same bug (#1050). Also removed the system OpenSSL dependency from the CLI, net –109 lines (#1038).

### hydra-db/hydradb

Rust | 2.3k stars | graph database

*5 open, 3 issues filed*

*Aug 2026 – Sep 2026*

- Wrote randomised differential tests over the sparse graph kernels. They disagree on a zero-hop start with no edges (#149, issue #148). The query optimizer returns different rows depending on which direction it expands the plan (#162), and synthetic Bolt relationship ids collide with the real id range (#163).

### rux-lang/Rux

Rust | 496 stars | systems language compiler

*5 merged of 7 opened*

*May 2026 – Jun 2026*

- The integer `**` operator crashed every program that used it (#34, 8 files). `\u{...}` escapes were dropped when lowering literals to HIR (#58). Constant integer expressions only coerced to sized int types as bare literals, so `1 + 1` and `2` typed differently (#72).

## PROJECTS

### kren | Rust | 5 stars

- Shared-memory IPC with no copies. One writer, many readers, a ring buffer in mapped memory, nothing serialised on the hot path.

### talos | Go, C, eBPF/XDP

- Layer 4 firewall. Filtering runs in the kernel at the XDP hook, before a packet reaches the network stack.

### aegis | Go

- A proxy that speaks the PostgreSQL wire protocol directly. Logs queries, routes reads away from writes, rewrites what it is allowed to and blocks what it is not.

## TECHNICAL SKILLS

**Languages:** Rust, C++23, Go, C, Python

**Systems:** SIMD (NEON, AVX2), aarch64 and x86-64 Linux, eBPF/XDP, mmap, wire protocols, mu-law audio

**Correctness:** differential and property testing, integer overflow and encoding boundaries, on-disk format stability

**Tooling:** Cargo, criterion, Docker, GitHub Actions, Linux perf

## EDUCATION

**Amrita Sai Institute of Science and Technology**

*B.Tech, Electronics and Communication Engineering*

Paritala, India

*Graduated 2026*